Energy resources

Finite fuels

1 Fossil & nuclear fuels

Coal, oil, gas and nuclear fuels are finite. Fossil fuels support heating, electricity and transport.

Burning fossil fuels releases CO_2 and pollutants. Nuclear: reliable electricity, low operational CO_2, radioactive waste.

2 Costs beyond fuel

Extraction damages habitats. Nuclear plants need expensive construction/decommissioning.

Biofuel can regrow, but uses land and releases CO_2 when burned. Carbon balance depends on regrowth and production.

Renewable supply

3 Wind, Sun & waves

No fuel combustion during operation, but output varies with weather. Solar also varies with daylight.

Wind: visual/noise/wildlife concerns. Waves: harsh conditions and maintenance. Solar: area and siting matter.

4 Water & Earth

Hydro reservoirs can supply on demand but flood land; drought reduces supply.

Tides are predictable, not continuous; estuary impacts. Geothermal is steady at suitable sites, with drilling costs.

Make a judgement

5 Match supply to use

Heating, transport and electricity have different needs. Use a mix, storage and flexible backup.

Renewable means replenished as used; it does not mean always available or impact-free.

6 Use evidence

Compare power, reliability, cost, emissions, waste and habitats. Justify which matters most.

Read totals and percentage shares carefully. Scientific evidence cannot alone settle political or economic choices.